

39 of 40 users have wanted to try a new category — and didn't.

Motivation is not the constraint. The app shows an unfamiliar category exactly what it shows a familiar one — a price. For a category you already buy that is enough; you hold the reference point in your head. For one you have never bought, a price with no reference point is uninterpretable, so evaluating it costs time a two-minute fetch session has not budgeted.

[Live MVP](#)[Repository & data](#)[Full written case study](#)

2 The barrier sits below the waterline, where features rarely look

EVENTS

React

"This user has bought the same two categories for six months."

PATTERNS

Anticipate

38/40 arrive knowing what they want. 20/40 crossed a category in 3 months — and their barriers did not soften.

STRUCTURE

Design

An unfamiliar category is shown the same information as a familiar one: a price, with no reference point.

MENTAL MODEL

Transform

User: "quick commerce is a fetch tool, not a browse tool." Company: "more recommendations produce more discovery."

Where solutions usually intervene

- A banner or an "explore" tab acts on EVENTS.
- A discount campaign acts on PATTERNS.
- Both push variables already non-zero — 39/40 want to explore.
- We act on STRUCTURE: supply the missing reference point.

The loop we are breaking

Narrow basket → faster session → "a fetch tool, not a browse tool" → no reference point for an unfamiliar price → less exploration → narrower basket

Vicious: each efficient session makes the next exploration less likely.

Fogg: $B = M \times A \times P$. Motivation 39/40 present · Prompt present · Ability is the binding constraint.

3

3,372 documents became 25 themes — every number computed, not generated

Collect

Normalize

Relevance
gate

Extract

Cluster

Synthesize

Validate

3,372

documents collected

Play Store 3,033 · App Store 339, across 5 quick-commerce apps

971

relevant — 28.8%

The gate removing 71% is the design: most reviews discuss delivery, not choice

25

interpretable themes

TF-IDF → LSA → agglomerative, merge threshold calibrated on homogeneity

0.4%

fabrication rate

4 in 924. Every quote must be a verbatim substring, asserted character-for-character

Three rules that make the output checkable

- The schema is the source of truth — every enum traces to one of the brief's 8 questions.
- Numbers never come from the model. It sees labels and quotes only.
- Contradicting evidence is computed, never requested.

The filter's error, found by interview

An interviewee described building their basket around a free-delivery threshold — a real constraint on what gets bought, expressed as a fee complaint. Re-checking the corpus: 133 documents use that language, and the gate discarded 64 as noise. Documented rather than quietly retuned.

Corpus is app-store review text only — Play Store 3,033 · App Store 339.

4 Trying to break our own insights: 3 pass, 2 report NOT RUN

Quote grounding

PASS

125 quotes checked against source text. 4 fabrications in 924 extractions; the rest are filtered before display.

Hold-out generalisation

PASS

Out-of-sample, against a size-matched permutation null. Real themes $z = +17.4$; word salad $z = -2.6$.

Theme non-degeneracy

PASS

An earlier run collapsed to one theme at 100% prevalence and reported success. Now asserted in code.

Source spread

NOT RUN

95% of the corpus is one source, so the check cannot discriminate. Reports the chance figure instead.

Inter-rater agreement

NOT RUN

Needs a hand-labelled gold set that does not exist. Reported absent, not approximated.

Cross-checked against a second model family

Re-run on Llama 3.3 70B and compared with Gemini across 25 themes: 0.66 Jaccard, no theme at zero overlap, mechanism text matching 12/25 vs a 4% baseline.

Agreement is not correctness — it bounds vendor dependence, nothing more.

Why report NOT RUN at all

Two of these used to print PASS over zero observations. A green line standing for no evidence is worse than an honest gap. Three pass, two abstain, and the deck says which.

5

Five interviews found three barriers 3,372 documents cannot contain

What the corpus missed	Found by	Why the corpus cannot hold it
"Comparing took too long, so I bought nothing"	Survey, 13/40	The user did not buy, so there is nothing to review. Engine sees it at 1.5% — a twentyfold gap, not sampling noise.
"This category does not apply to me"	Interviews, 3 of 5	No need means no purchase means no review. All 8 barrier codes presume latent demand, so all three would be miscoded as blocked.
"I build my basket around the free-delivery threshold"	Interview + 133 corpus docs	Present in the text at scale — but the gate read it as a delivery complaint and the schema has nowhere to store it.

A corpus of reviews contains only people who completed a transaction. Every barrier that stops someone before they buy generates no text at all.

Where research CONFIRMED the engine

Barrier ranking matches across two independent datasets. Engine: trust/quality 45.3%, price risk 22.7%, awareness 4.8% — the survey ranks them identically.

Where the engine outran both

Its third-largest barrier is return anxiety at 18.4% — barely probed by the survey, raised by no interview. The correction runs both ways.

6 Every individual skip is rational; the aggregate is a two-year rut

Target segment

- ≥ 3 orders/month (27 of 40)
- ≥ 6 months tenure
- ≤ 2 categories carry most spend
- Already acquired and transacting — the blocker is behavioural.

Root cause

The app presents unfamiliar categories with the same information as familiar ones. For a familiar category that is enough, because the user supplies the missing context from memory. For an unfamiliar one it is not — so evaluating it costs time the user has not budgeted, and the rational move inside a two-minute fetch session is to skip it.

Every skip is individually correct. The aggregate outcome is a user who has bought the same things for two years.

Existing workarounds — people only build these for problems they actually have

Buy it somewhere else

D-Mart 8 · Kirana 6 · Amazon 3

The demand is not missing — it is leaking. Displaced revenue, not absent appetite.

Wait for a price signal

“cheaper elsewhere” 12 · “too expensive” 8

Price is being used as a proxy for risk. Cheap enough means safe enough to test.

Validate elsewhere

14 want reviews first

The evaluation happens — on Instagram or in a shop. And so does the purchase.

Defer indefinitely

“forgot once cart-filling” 3 · distracted 6

Intent exists but evaporates on contact with the reorder flow.

Where the model stops: 2 of 5 interviewees have barriers no information closes — an incumbent kirana, and a belief settled by one bad delivery. The MVP shows them nothing, which is the correct behaviour.

7 One lever: cut evaluation cost at cart review

BUSINESS OUTCOME

Retention & LTV — category breadth is the moat

NORTH STAR METRIC

CER — % of MAC buying from ≥ 1 new category this month

THE DEFINITION THAT DECIDES IT

A purchase in month M from a category bought in none of M-1 ... M-6.

CER = eligible sessions × surfaced × considered × converted

Eligible sessions

Shopper reaches cart review with ≥ 2 orders of history

Instrumented, not modelled

Surfaced

% of those sessions where a substantiated new category exists

35 of 36 row entries would register in CER

Considered

1 – dismissal rate

Pre-registered guardrail: kill above 60%

Converted

Add-to-cart on a suggested new category

The number the experiment moves

Opportunity → the one thing we build

Three of the four are already healthy: intent arrives in-app (Blinkit 16 vs Amazon 3), sessions are frequent, the catalogue is there. Only CONSIDERED is not — the shopper cannot evaluate fast enough. That is what the intervention targets.

The correction that changed the build

The spec first ranked the repeat prompt above first trial, reasoning that a monthly metric rewards repeats. It does not — that repeat sits inside the lookback and scores zero. First crossovers now lead.

8

The MVP pre-computes the comparison a two-minute session cannot afford

User flow — no new screen, no new journey

1

Fetch-mode session

Shopper opens the app knowing what they want — 38/40

2

Basket fills

Their usual list. The agent does not interrupt the task

3

Cart review

The only natural pause: task done, payment not started

4

Card appears

Up to 5, each opening a different never-bought category

5

Add, or dismiss

Dismissal is logged as a guardrail, not hidden

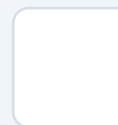
Every line on the card closes a blocker the survey named

Reason from own history → “not sure it’ll suit my need” 14/40 · Price anchor → “priced higher than usual” 15/40 · Trust signal → “don’t trust the brand / want reviews” 26/40 · Smallest pack → “too expensive” 20/40

Wireframe — the card, at cart review

CATEGORY SPARK

5 CATEGORIES

**Drools Puppy Treats · 150g****₹65**

“You have a pet — pet supplies is the usual next thing people add”

~₹65/week · **your usual runs**

~₹68/week

Reason

their own history, never “customers like you”

Price anchor

₹/week vs their own comparable spend

Trust signal

most-reordered pick in that category

Add to Cart

[Try the live MVP — 10 shopper profiles, two of which correctly receive no suggestion at all](#)

9 Decisions are deterministic — the LLM only writes the sentence

1 Eligibility

Mode A first crossover · Mode B
lapsed repeat · or neither

*Rules, no model call.
Fails closed to no
card.*

2 Profile

Order history → owned
categories, cadence, declared
signals

*Read from the
shopper's own data*

3 Candidates

Never-bought categories, minus
any the shopper has closed

*Also the only list the
LLM may choose
from*

4 Price anchor

₹/week for the candidate vs
their own comparable spend

*Division. Never
routed through a
model*

5 Compose

One reason line + one trust line

*The only step an
LLM touches*

No API key required

Mode, category, product and the anchor are identical with or without an LLM — only the wording differs. Copy is pre-generated in CI where the key lives, and discarded at render if the agent picked differently than it did at build time. A demo that a quota reset can break is not a demo.

Agent evals — 11 pass, 1 not run

- Never an owned or closed category
- Every SKU exists and is in stock
- Every reason traceable to history or a declared signal
- The rendered ₹/week recomputes exactly
- Each check is mutation-tested

Deployed via GitHub Actions to GitHub Pages. The deploy job runs the evals first and refuses to publish on failure.

10 Pre-registered guardrails, and what would kill this

Guardrails — committed in advance

- No AOV degradation
- No drop in core-category order frequency — >3% is a kill
- No rise in refund / return rate
- Card dismissal rate below 60%
- Decided before the result, so it cannot be relitigated after

What would falsify the hypothesis

- Cards are added but the category never repeats — trial without repeat is a discount, not exploration
- Dismissal climbs with exposure — the card is noise
- CER rises while core frequency falls — we moved attention, not behaviour

Limits we state rather than bury

- n=40 convenience sample, skewed young and metro. Baby (3) and pet (2) owners too thin for segment claims
- n=5 interviews: enough to challenge the vocabulary, never enough for an “X% of users” claim — and none is made
- The MVP runs on synthetic order histories. It demonstrates the mechanism, not real-world lift

No interview was fabricated, and no synthetic persona is counted toward the total. Of five hypotheses the AI personas produced, one was testable — and it was wrong.

[Live MVP](#) · [Repository, corpus and evals](#) · [Full written case study](#)